

Exam solution 2025

Question 2.1

One way to determine whether these bonds present an arbitrage opportunity, we can check whether the pricing is internally consistent. For example, calculate the rates implied by Bonds A, B, and D (the zero-coupon bonds), and use this to check Bond C.

$$934.58 = 100 / (1 + YTM_1), YTM_1 = 7\%$$

$$881.66 = 100 / (1 + YTM_2)^2, YTM_2 = 6.5\%$$

$$839.62 = 100 / (1 + YTM_3)^3, YTM_3 = 6\%$$

Given the spot rates implied by Bonds A, B, and D, the price of Bond C should be \$1,105.21. Its price really is \$1,118.21, so it is overpriced. Yes, there is an arbitrage opportunity.

To take advantage of this opportunity, you want to (short) Sell Bond C (since it is overpriced). To match future cash flows, one strategy is to sell 10 Bond Cs (it is not the only effective strategy; any multiple of this strategy is also arbitrage). This complete strategy is summarized in the table below.

	Today	1 Year	2Years	3Years
Sell Bond C	11,182.10	-1,000	-1,000	-11,000
Buy Bond A	-934.58	1,000	0	0
Buy Bond B	-881.66	0	1,000	0
Buy 11 Bond D	-9,235.82	0	0	11,000
Net Cash Flow	130.04	0	0	0

Note that to get full score it is essential to replicate a strategy with *no risk* and comment on that.

Question 2.2

$$\text{Price} : 820 / 1.07^4 = 625.574$$

$$\text{Yield to maturity: Solution of } 820 / 1.07^4 = 1000 / (1 + YTM)^4 \text{ for } YTM = 12.44\%$$

The yield to maturity of a corporate bond is based on the promised payments of the bond. But there is some chance the corporation will default and pay less. Thus, the bond's expected return is typically less than its YTM.

Corporate bonds have credit risk, which is the risk that the borrower will default and not pay all specified payments. As a result, investors pay less for bonds with credit risk than they would for an otherwise identical default-free bond. Because the YTM for a bond is calculated using the promised cash flows, the yields of bonds with credit risk will be higher than that of otherwise identical default-free bonds. However, the YTM of a defaultable bond is always higher than the expected return of investing in the bond because it is calculated using the promised cash flows rather than the expected cash flows.

Question 3.1

- **Project A – Emissions Reduction Initiative**
Focus: Upgrade of production equipment
→ It clearly targets **CO₂** and **eutrophying emissions**, not working conditions in the supply chain.
- **Project B – Ethical Supply Chain Program**
Focus: Restructuring the supply chain, audits, and community training
→ This is the one that is naturally linked to **eliminating the 0.5 FTE of forced labour**.
- **Project C:** no quantified evidence it meets targets

Implication: under strict sustainability constraints (if we must hit *all* three targets), none of the project should be prioritized even though they are all positive NPV projects. Any reasonable reflection on constraints is ok.

Question 3.2

Discount each year's 20 M€ at 5% and sum:

Year 1: $20 / 1.05 \approx 19.05$; Year 2: $20 / 1.05^2 \approx 18.14$; Year 3: $20 / 1.05^3 \approx 17.28$; Year 4: $20 / 1.05^4 \approx 16.45$; Year 5: $20 / 1.05^5 \approx 15.67$

Sum of discounted cash flows: ≈ 86.59 M€

Financial NPV: $86.59 - 60 = 26.59$ M€

2. PV of sustainability benefits

Annual euro benefit:

CO₂: $100\,000 \text{ kg} \times 0.163 \text{ €/kg} = 16\,300 \text{ €} = 0.0163 \text{ M€}$

P: $750 \text{ kg} \times 209 \text{ €/kg} = 156\,750 \text{ €} = 0.1568 \text{ M€}$

Total per year:

$0.0163 + 0.1568 \approx 0.1731 \text{ M€/year}$

Discount this 5 years at 2%: PV of sustainability benefits ≈ 0.79 M€

3. Integrated NPV for Project A

26.59 (financial) + 0.79 (sustainability) ≈ 27.38 M€

Project B – Ethical Supply Chain Program

Discount each year's 25 M€ at 5%: $(25 / 1.05) + (25 / 1.05^2) + \dots + (25 / 1.05^5) \approx 108.24$ M€

Financial NPV: $108.24 - 70 \approx 38.24$ M€

2. PV of sustainability benefits

Annual euro benefit from forced labour reduction:

$0.5 \text{ FTE} \times 123\,000 \text{ €/FTE} = 61\,500 \text{ €} = 0.0615 \text{ M€/year}$

Discount 5 years at 2%: ≈ 0.28 M€

3. Integrated NPV for Project B

$38.24 \text{ (financial)} + 0.28 \text{ (sustainability)} \approx 38.52 \text{ M€}$

Project C – Circular Economy & Recycling

Discount each year's 22 M€ at 5%: $(22 / 1.05) + \dots + (22 / 1.05^5) \approx 95.25 \text{ M€}$

Financial NPV: $95.25 - 50 \approx 45.25 \text{ M€}$

2. PV of sustainability benefits

Annual euro benefit:

$\text{CO}_2: 50\,000 \times 0.163 = 8\,150 \text{ €} = 0.00815 \text{ M€}$

$\text{P: } 375 \times 209 = 78\,375 \text{ €} = 0.07838 \text{ M€}$

Total per year:

$0.00815 + 0.07838 \approx 0.0865 \text{ M€/year}$

Discount 5 years at 2% $\approx 0.40 \text{ M€}$

3. Integrated NPV for Project C

$45.25 \text{ (financial)} + 0.40 \text{ (sustainability)} \approx 45.65 \text{ M€}$

So under this integrated NPV approach:

Project C has the highest integrated NPV and should be prioritized if the firm's only objective is to maximize the sum of financial and monetized sustainability value.

Could different strategic priorities justify a different choice?

Yes. So far we assumed that those monetization factors fully capture the firm's (or society's) valuation of impacts, and all dimensions are perfectly substitutable in euros.

But in reality, if eliminating forced labour is a non-negotiable ethical requirement or a reputational “must have”, the firm might give lexicographic priority to Project B, even though its integrated NPV is lower than C's.

If regulators or investors are focused on meeting specific climate and eutrophication targets, they might prefer Project A, which delivers the largest environmental reductions, even though its integrated NPV is lowest.

A firm might also use integrated NPV as a screening tool, then apply additional strategic or ethical criteria.

Question 3.3

- **Monetizing environmental and social impacts**
 - **Why it occurs:** Many impacts (CO₂, ecosystem damage, forced labour, human health) have no clear market price. Monetary values must be based on modelling, surveys, or policy choices, which are uncertain and normative.
 - **Effect on results:** Integrated NPVs become highly sensitive to the chosen monetization factors. Rankings can flip if these values change, and the final numbers may give a false impression of objectivity and precision.
- **Choice of time horizon and discount rates**
 - **Why it occurs:** Financial cash flows and long-term sustainability impacts (e.g. climate, biodiversity, social outcomes) operate on different time scales. There is no consensus on what discount rate is appropriate for impacts affecting future generations.
 - **Effect on results:** Small changes in the discount rate can strongly change the present value of long-term impacts and alter which project appears “best.” This can systematically bias decisions toward short-term financial gains and make the interpretation of “optimal” projects dependent on contested ethical assumptions about the future.

Question 4.1

Beta

$$\text{Cov} = ((0.08 - 0.04)(0.1 - 0.06) + (-0.02 - 0.04)(-0.01 - 0.06) + (0.06 - 0.04)(0.09 - 0.06))/3 = 0.00213333$$

$$\text{VarM} = ((0.08 - 0.04)(0.08 - 0.04) + (-0.02 - 0.04)(-0.02 - 0.04) + (0.06 - 0.04)(0.06 - 0.04))/3 = 0.00186667$$

$$\text{Beta_Eo} = \text{Cov}/\text{VarM} = 1.14286$$

rU, rE, rD

$$r_{Do} = 0.03 + 0.05(0.04 - 0.03) = 0.0305$$

$$r_{Eo} = 0.03 + \text{Beta_Eo}(0.04 - 0.03) = 0.0414286$$

$$r_U \text{ Solution of } r_{Eo} = r_U + 0.2(r_U - r_{Do})$$

$$r_U = 0.0396071$$

$$r_{Da} = 0.03 + 0.1(0.04 - 0.03) = 0.031$$

$$r_{Ea} = r_U + 0.7(r_U - r_{Da}) = 0.0456321$$

Alternative solution

$$\beta_U \text{ is a solution of } \beta_{Eo} = \beta_U + 0.2(\beta_U - 0.05) \text{ or } \beta_U = \frac{1}{1+0.2}\beta_{Eo} + \frac{0.2}{0.2+1}0.05$$

$$\beta_U = 0.960714$$

$$\beta_{Ea} = \beta_U + 0.7(\beta_U - 0.1)$$

$$1.56321$$

$$r_U = 0.03 + \beta_U(0.04 - 0.03) = 0.0396071$$

$$r_{Ea} = 0.03 + \beta_{Ea}(0.04 - 0.03) = 0.0456321$$

WACC

$$r_{WACC} = r_{Ea} \frac{1}{1 + 0.7} + r_{Da} \frac{0.7}{0.7 + 1} (1 - 0.22) = 0.0367989$$

Question 4.2

$$V_U = 14/(r_U - 0.02) = 714.026$$

$$V_L = 14/(r_{WACC} - 0.02) = 833.388$$

$$\text{Tax shield} = V_L - V_U = 119.362$$

$$\text{Or explicitly } (0.22 r_{Da} \frac{0.7}{0.7 + 1} V_L)/(r_U - 0.02) = 119.362$$

Question 4.3

For D/E=0.2

$$r_{Ea2} = r_U + 0.2 (r_U - r_{Da}) = 0.0413286$$

$$r_{WACC2} = r_{Ea2} \frac{1}{(1 + 0.2)} + r_{Da} \frac{0.2}{(0.2 + 1)} (1 - 0.22) = 0.0384705$$

$$VL2 = 14 / (r_{WACC2} - 0.02) = 757.966$$

For D/E=0.4

$$r_{Ea4} = r_U + 0.4 (r_U - r_{Da}) = 0.04305$$

$$r_{WACC4} = r_{Ea4} \frac{1}{(1 + 0.4)} + r_{Da} \frac{0.4}{(0.4 + 1)} (1 - 0.22) = 0.0376586$$

$$VL4 = 14 / (r_{WACC4} - 0.02) = 787.816$$

Then we have that D/E=0.4 maximizes the firm value

$$\{VL2, VL4, VL - 50\} = \{757.966, 787.816, 783.388\}$$

When managers worry about “bankruptcy costs,” they usually mean **both direct and indirect costs** of financial distress.

1. Legal and administrative costs in restructuring or bankruptcy

- Lawyers, court fees, advisors, auditors, restructuring consultants, and administrative costs of formal proceedings (reorganization, Chapter 11–type processes, etc.).
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2. Loss of business due to damaged reputation and counterparty distrust

- Customers may switch to more stable competitors because they fear delays, non-performance, or poor service.
- Suppliers may tighten trade credit terms (demand cash up front, shorten payment periods) or stop delivering altogether.

3. Loss of key employees and underinvestment in the business

- Talented managers and specialists may leave a financially distressed firm because of uncertainty about its future, making it harder to operate efficiently.

